FRENKY RISKI GILANG PRATAMA

Machine Learning & AI Enthusiast
Data Analysis | Deep Learning & Computer Vision
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PROFESSIONAL SUMMARY

Analytical and detail-oriented Mathematics graduate from Ahmad Dahlan University with a GPA of 3.65/4.00 and a strong focus on Data Analysis, Machine Learning, and Artificial Intelligence. Proficient in Python, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, and PyTorch, with hands-on experience in predictive modeling, CNN-based image classification, LLM integration, and interactive data visualization. Demonstrated research experience, including a deep learning model achieving 99.79% test accuracy, alongside expertise in model evaluation and reproducible experimentation. Passionate about transforming complex data into actionable insights and developing reliable, data-driven AI solutions.

EDUCATION

Ahmad Dahlan University - Yogyakarta, Indonesia **GPA: 3,65/4,00**
Bachelor of Mathematics *September 2022 - August 2026*

- **Relevant Coursework:** Statistics, Probability, Linear Algebra, Calculus, Numerical Analysis, Discrete Mathematics, Mathematical Modeling, Mathematical Analysis.

PROFESSIONAL EXPERIENCE

PSDS Mentor - Advanced Class *2025*

- Developed and delivered CNN learning materials in notebook format, covering data augmentation, convolution, pooling, activation functions, fully connected layers, model training, and evaluation.
- Developed hands-on modules using Jupyter Notebook and PyTorch to guide 20+ participants in CNN implementation and training and evaluation pipelines.
- Facilitated CNN practice through nine thematic notebooks, covering model training with and without data augmentation.
- Evaluated model performance using confusion matrices, classification reports, and classification metrics to analyze prediction results.

Machine Learning - Bangkit Academy 2024 Batch 2 *2024*

- Completed a Machine Learning learning path covering Python, data analysis, mathematics for machine learning, TensorFlow, generative AI, and model deployment.
- Developed a facial recognition model for user authentication and identification as part of the OrganiSync capstone project.
- Contributed to OrganiSync, selected among the Top 50 Product-Based Capstone Projects out of 644 projects.
- Achieved Full Graduate status upon completing the program's learning activities and project requirements.

PSDS Excel Class Mentor *2024*

- Taught Microsoft Excel, PivotTables, mathematical formulas, and data-processing functions to support efficient data analysis.
- Trained participants in Excel shortcuts, data formatting techniques, and spreadsheet features.
- Guided 50+ participants in creating charts and data visualizations to present analytical findings effectively.
- Designed and delivered practical case studies to help participants apply Excel in academic and professional settings.

AI & DATA PROJECTS

Twitter Spam Detection - IBM SkillsBuild 2025

- Developed a text classification project using crawled Twitter/X data to analyze spam patterns in digital conversations.
- Applied the IBM Granite LLM to classify text as Spam or Non-Spam and generate automated explanations for classification results.
- Implemented text summarization to highlight key information and insights from text datasets.
- Built an interactive dashboard using Streamlit and WordCloud to visualize spam distribution, language patterns, and analytical results.

RESEARCH & PUBLICATIONS

Hybrid CNN and LLM for Image-Based Classification of Plant Leaf Diseases
First Author | Published August 21, 2026

- Developed a plant leaf disease classification model using EfficientNetV2-M and GPT-4o to generate diagnostic explanations.
- Achieved 99.79% test accuracy through deep learning model development and evaluation.
- Integrated an LLM to explain symptoms, distinguish disease characteristics, and suggest preliminary actions, supporting explainable AI in digital agriculture.
- Published in International Journal of Advances in Data and Information Systems (IJADIS), Vol. 7, No. 2 (2026), pp. 573–584 (SINTA 2).

Comparative Analysis of CNN Architectures and Optimizers for ALL Classification
Third Author | Published August 23, 2026

- Developed a PyTorch-based image classification pipeline for Acute Lymphoblastic Leukemia (ALL), comparing six CNN architectures and four optimizers.
- Implemented preprocessing, data augmentation, transfer learning, and model training and testing.
- Evaluated performance using accuracy, loss, confusion matrix, precision, recall, and F1-score.
- Designed experimental pipelines and saved top models for standardized, reproducible evaluation.
- Contributed to manuscript preparation, including methodology, experiments, results discussion, and interpretation of evaluation metrics.

ORGANIZATIONAL & LEADERSHIP EXPERIENCE

General Secretary - Karang Taruna Permata Muda Dusun Siluk 1 *February 2024 - Present*

- Managed 50+ organizational records, including correspondence, meeting minutes, and activity documents.
- Coordinated meetings, information sharing, decisions, task assignments, and follow-ups.

Head of Human Resource Development - Student Executive Board (BEM) *June 2024 - May 2025*

- Analyzed student development needs through surveys to inform effective program planning.
- Designed soft-skill development programs and managed a database of [100+] committee members from BEM and student associations (HIMA).

Event Chair - RESPECT 13 *October 2023*

- Planned key event components, including the concept, theme, schedule, budget, and proposal.
- Coordinated faculty, vendors, and committees to deliver events on schedule and within budget.

Facilitator Coordinator - Faculty of Applied Science and Technology Campus Introduction Program *September 2024*

- Coordinated facilitators in guiding new students, providing orientation, and answering academic and non-academic questions.
- Resolved student concerns and supported campus adjustment, discipline, ethics, and character development.

CERTIFICATIONS & TRAINING

- **Data Classification and Summarization Using IBM Granite - IBM SkillsBuild** *2025*
  Large Language Model (LLM), Text Classification, and Data Summarization
- **PSDS Advanced Class** *2024*
  Machine Learning, Deep Learning, Image Classification, and Data Analysis
- **Bangkit Academy 2024 Batch 2 - Machine Learning Learning Path** *2024*
  Python, Data Analysis, Machine Learning, TensorFlow, and Generative AI
- **PSDS Elementary Class** *2023*
  Python, SQL, Database Design, and Data Analysis

SKILLS

Languages: Indonesian (Native), English (Intermediate).

Technical Tools: Python, SQL, PyTorch, TensorFlow, Scikit-learn, Keras, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Google Colab, Visual Studio Code, RStudio, DBeaver, Microsoft Excel, Streamlit.

Hard Skills

- **Data Analysis:** Data cleaning, exploratory data analysis, statistical analysis, and result interpretation.
- **Machine Learning:** Feature engineering, model training, hyperparameter tuning, and model evaluation.
- **Deep Learning:** CNNs, transfer learning, data augmentation, image classification, and PyTorch.
- **AI Engineering:** LLM integration, generative AI, explainable AI, model deployment, and interactive dashboards.

Soft Skills: Analytical Thinking, Critical Thinking, Problem-Solving, Communication, Leadership, Teamwork, Collaboration, Decision-Making, Adaptability, Time Management, Attention to Detail, Teaching & Mentoring.

ADDITIONAL INFORMATION

- **Career Interests:** Data Analyst, Machine Learning Engineer, AI Engineer, Data Scientist, Deep Learning Engineer.